Graph Colouring

Theorem 1

A graph G is bipartite $\iff \chi(G) = 2$.

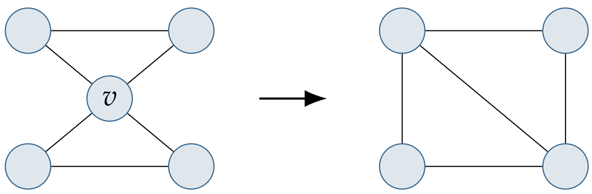
Proof. ($\Rightarrow$) Since G is bipartite, the vertex set can be split into two sets A and B where all edges connect vertices from different sets. Assign colour 1 to the vertices of A and colour 2 to the vertices of B; this yields a 2-colouring. Clearly a bipartite graph is not 1-colourable (unless it has no edges).

($\Leftarrow$) Since $\chi(G) = 2$, we construct two sets A, B based on colour: put the vertices of colour 1 in set A, and the vertices of colour 2 in set B. Two vertices from the same set are not adjacent, since they have the same colour. Hence G is a bipartite graph. □

Theorem 2

If G is a simple graph with largest vertex degree Δ, then G is $(\Delta + 1)$-colourable.

Proof. The proof is by induction on the number of vertices of G. Let G be a simple graph with n vertices. If we delete any vertex v and its incident edges, the graph that remains is a simple graph with $n - 1$ vertices and largest vertex degree at most Δ. By the induction hypothesis, this graph is $(\Delta + 1)$-colourable. A $(\Delta + 1)$-colouring for G is then obtained by colouring v with a colour different from the vertices adjacent to v.



□

Theorem 3

Let G be a connected graph whose maximum vertex degree is d. If G is neither a cycle graph with an odd number of vertices nor a complete graph, then $\chi(G) \leq d$.

Remark. This is Brooks' Theorem. (No proof was included in the original notes.)

Theorem 4

The vertices of any simple connected planar graph G can be coloured with six (or fewer) colours.

Proof. We prove the theorem by induction on the number of vertices. The result is trivial for simple planar graphs with at most 6 vertices. Suppose then that G is a simple planar graph with n vertices, and that all planar graphs with $n-1$ vertices are 6-colourable.

We know that G contains a vertex v of degree at most 5; if we delete v and its incident edges, then the graph that remains has $n-1$ vertices, and is thus 6-colourable. Then a 6-colouring of G is obtained by colouring v with a colour different from the vertices adjacent to v. □

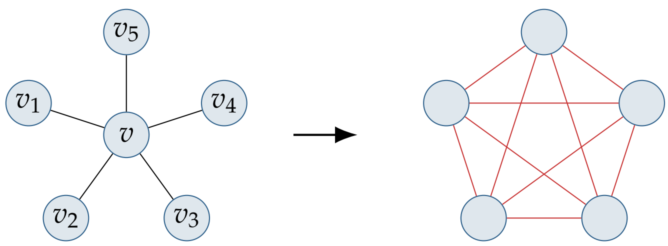
Theorem 5

The vertices of any simple connected planar graph can be coloured with five (or fewer) colours.

Proof. Induction on the number of vertices. The result is trivial for simple planar graphs with fewer than six vertices, and we assume that all simple planar graphs with $n-1$ vertices are 5-colourable. G contains a vertex v of degree at most 5. As before, the deletion of v leaves a graph with $n-1$ vertices, which is thus 5-colourable. Our aim is to colour v with one of the five colours.

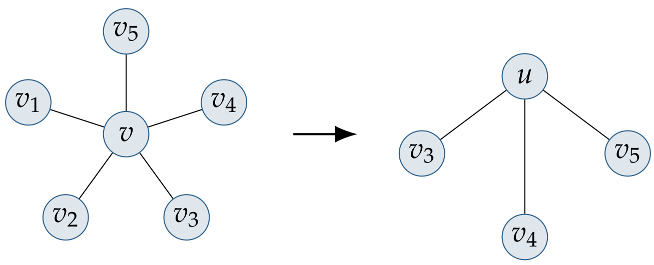
If $\deg(v) < 5$, then v can be coloured with any colour not assumed by the (at most four) vertices adjacent to v, completing the proof.

If $\deg(v) = 5$, and the vertices $v_1, \ldots, v_5$ adjacent to v are arranged around v in clockwise order: if the vertices are mutually adjacent, then G contains the non-planar graph K_5 as a subgraph, which is impossible.



K_5 (contradiction) as a subgraph

So at least two of the vertices v_i are not adjacent (say v_1 and v_2). We contract the two edges vv_1 and vv_2; the resulting graph is a planar graph with fewer than n vertices, and is thus 5-colourable.



We now reinstate the two edges, giving both v_1 and v_2 the colour originally assigned to v. A 5-colouring of G is then obtained by colouring v with a colour different from the (at most four) colours assigned to the vertices v_i. □

Theorem 6 (Four Colour Theorem)

The vertices of any simple connected planar graph can be coloured with four (or fewer) colours.

Remark. Stated without proof in the original notes (the theorem's known proof is computer-assisted).